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02 Logical operator

Logical operator are used to combine multiple condition and return either true @ false

There are 3 logical operators-

1. And :- Both condition need to be true then its true
2. OR :- Any one condition needs to be true
3. Not :- vice versa

Ex:- a = 10
b = 15

```
print (a==b) and (b!=a)
print (a==b or b!=a)
print (a!=b)
```

output :- False
True
True

03 Comparison operator / relational operator

comparison operator are used to compare two values, they always return in Boolean (True, false)

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Comparison operators are

1. ==
2. !=
3. >
4. <
5. >=
6. <=

Ex:- a =10
b =15

```
print(a > b)
print (a < b)
print (a == b)
print (a!= b)
print (a >= b)
print (a <= b)
```

output:- False
True
False
True
False
True

04 Assignment operators

Assignment operators are used to assign values to variables.

Assignment operators are:-

+= (Add & assign)
-= (Sub & assign)
*= (Multiply & assign)
/= (divide & assign)
//= (floor divide & assign)
%= (Modulus & assign
**= (power & assign)

Ex:- a = 10
b = 15

```
a += b  # a = a + b
print (a)

a -= b  # a = a - b
print (a)

a *= b  # a = a * b
print (a)

a /= b  # a = a / b
print (a)

a //= b  #a = a // b
print (a)
a **= b  # a = a ** b
print (a)
```

```
a %= b  # a = a % b
print (a)
```

output :- 25
10
150
10.0
0.0
0.0
0.0

05 Bitwise operator

Bitwise operator are used to perform operations on the "Binary" (bit-level) representations of numbers.

* In computer every number is stored in 0's & 1's
* Bitwise operator allow us to work directly on this basis

<u>Binary language</u>

16 8 2 1

$2^3(8)$ $2^2(4)$ $2^1(2)$ $2^0(1)$

$2^3(8)$	$2^2(4)$	$2^1(2)$	$2^0(1)$	
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8

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Ex:- 11 & 12

↓ ↓

1011 1100

```
 1011
 1100
 ----
 1000 => 8
```

∴ ~~11 & 12~~ 11 & 12 = 8

types of Bitwise operators are:-

1) And (&)

a = 5

b = 3

print (a & b)

```
 0101
 0011
 ----
 0001 = 1
```

2) OR (|)

print (5 | 3)

```
 0101
 0011
 ----
 0111 = 7
```

3) XOR (^)

print (5 ^ 3)

0101
0011
0110 = 6

4) NOT (~)

print (~5)

output = 6

5) Lift shift (<<)

print (5 << 1)

0101 -> 1010
5 becomes 10

6) Right shift (>>)

print (5>>1)

0101 = 0010
5 becomes2

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06 Identity operators

Identity operators are used to check whether two variables refer to the same object in memory.

1) is
2) is not

Ex:- a = [1, 2, 3]
b = a
print (a is b)

output b = [1, 2, 3]

a = 8
b = 8
print (a is b)
print (a is not b)

output :- True
False

07 Membership operators

Membership operators are used to check whether a value exist inside a sequence

A sequence can be list, tuple, string, set etc...

01) in
02) not in

Ex:- l1 = [4, 1, 3, 5]
print (5 in l1)
print (3 not in l1)

output:- ~~False~~ True
False

Example problem 03

write a program to calculate, area of square, area of circle, area of triangle take input from users.

```
=> s = float (input ("enter square side: "))
r = float (input ("enter the radius of circle: "))
b = float (input ("enter the base of triangle: "))
h = float (input ("enter the height of triangle: "))

area_square = s * s
~~area~~ radius_circle = 3.14159 * r * r
area_triangle = 0.5 * b * h

print ("Area of square")
print (" Area of circle)
print (" Area of triangle)
```